In Scope:

- Track and log driver hours of service (HOS) against regulatory limits, with violation warnings
- Real-time fatigue detection alerts based on driving hours, time-of-day, and rest patterns
- Route safety analysis using conditions data (e.g., weather, road/traffic risk) to flag high-risk routes
- Centralized dashboard for fleet managers to view driver status, alerts, and route risk in one place
- Historical reporting on HOS compliance, fatigue incidents, and route risk trends
- Role-based access (driver view vs. fleet manager/admin view)

Out of Scope:

- Physical fatigue-sensing hardware (e.g., eye-tracking cameras, biometric sensors) — system will rely on logged hours/patterns and input data, not raw biometric monitoring
- Integration with real commercial ELD (Electronic Logging Device) hardware or live GPS telematics feeds — project will use simulated/sample data
- Legal certification as an official FMCSA-compliant ELD replacement
- Payroll, dispatch/routing optimization, or vehicle maintenance tracking features
- Mobile app deployment to app stores (prototype/web-based demo only, for class scope)